

RYAN KIM

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U.S. Citizen

Open to relocation and nationwide travel opportunities

TECHNICAL SKILLS

Mechanical Design: SolidWorks; Siemens NX; GDT; Tolerance Stack-Ups; DFM; DFA; FEA; 2D Drawings; BOM Management

Manufacturing: Prototype Assembly; Fixture Design; Root Cause Analysis; ISO 9001

Laboratory: Mechanical Testing; Failure Analysis; Data Analysis; Instron Testing; Optical Microscopy; GLP

Programming: Python; MATLAB; C++

EXPERIENCE

Volta Space Technologies

April 2026 – July 2026

Mechanical Design Intern

Broomfield, CO

- Developed SolidWorks CAD models and mechanical hardware for internal and electro-optical support applications.
- Designed optical-mount fixtures and supported hardware development for future thermal vacuum (TVAC) testing.
- Supported engineering documentation, BOM management, and technical reporting using Microsoft 365.
- Participated in tolerance analysis, design iteration, and system-integration discussions.

Robomechanics

Nov 2025 – Feb 2026

Mechanical Engineer (Contract, 3 months)

Teterboro, NJ

- Converted legacy metric CAD assemblies into inch-based production models and ASME Y14.5-compliant drawings for pallet-handling robotics within an ISO 9001 quality program.
- Performed tolerance stack-ups on sheet-metal and welded assemblies to resolve fit-up, interference, and failure issues.
- Worked with machinists and fabricators to revise geometry and interfaces around manufacturing constraints and shop-floor feedback.
- Designed welding and assembly fixtures to control distortion, maintain alignment, and improve fabrication repeatability.
- Developed fixture strategies around part geometry and weld sequencing to improve fit-up and reduce prototype rework.

Lamont-Doherty Earth Observatory

Summer 2021

Research Assistant

Palisades, NY

- Supported research on microplastic contamination in consumer detergents through sample preparation, filtration, and laboratory testing.
- Performed UV-Vis spectrophotometry, followed cleanroom procedures, analyzed data, and documented preliminary findings.

EDUCATION

Northeastern University

Boston, MA

M.S. Mechanical Engineering, Concentration in Materials, Incoming Fall 2026

Sep 2026 – May 2028

Rutgers University, School of Engineering

New Brunswick, NJ

B.S. Biomedical Engineering

Sep 2021 – May 2025

- Coursework in biomaterials, biomechanics, and biomedical devices focused on the mechanical, biological, and chemical behavior of engineered systems.
- Performed mechanical testing, failure analysis, and data analysis using Instron machines and optical microscopy.

PROJECTS

View full project portfolio: rkydin.github.io

ParaSwing - Wheelchair-Mounted Golf Swing Assist Device

Sep 2024 – May 2025

Senior Capstone Design Project

- Led the SolidWorks mechanical design of a wheelchair-mounted robotic golf swing assist device, including structural parts, mounting hardware, and motion-transfer mechanisms.
- Fabricated and iterated prototypes using additive manufacturing and electromechanical components, then integrated controls and a mobile interface.